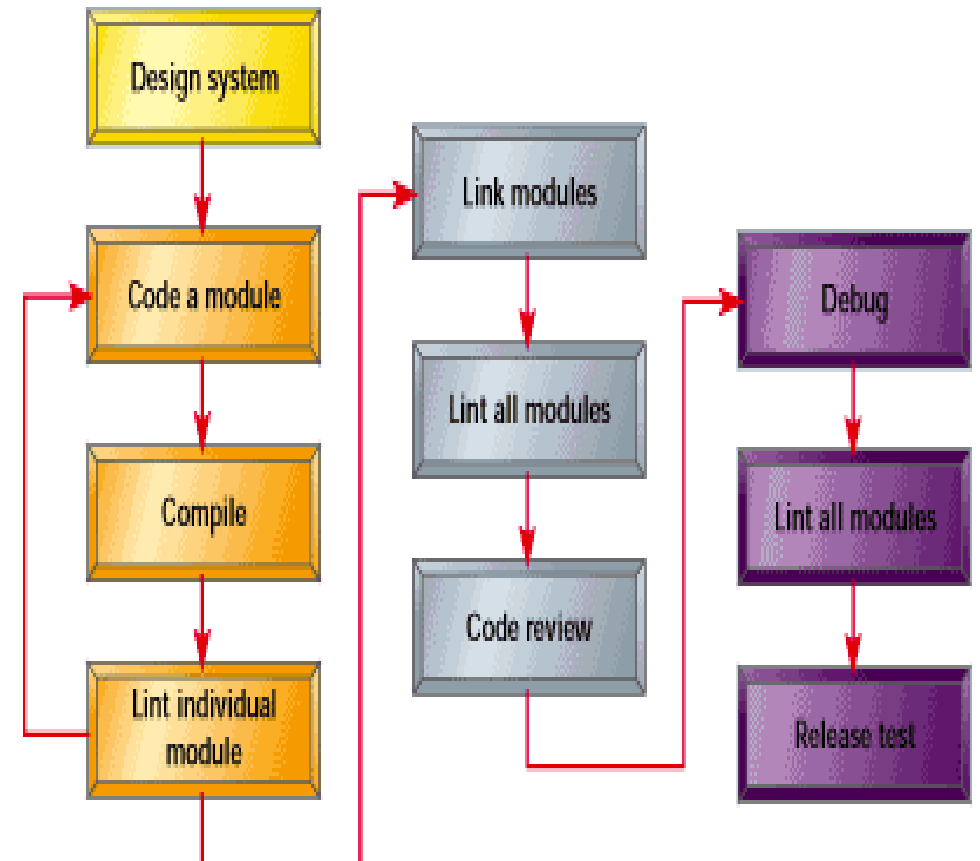


Lint and Valgrind

C programming tools for static analysis and memory management

Lint

- **Lint**, or a **linter**, is a tool that analyses **source code** to flag programming errors, **bugs**, stylistic errors, and suspicious constructs.
- Possible indexing beyond array bounds
- De-referencing of null pointers
- Suspicious assignments (such as if (a = b))
- Mismatches in variable types (such as foo declared as a double in one file and used as a long in another)
- Unused variables
- Unreachable code
- Header files included multiple times and/or unnecessarily



~cs537-1/projects/lint/cpplint.py

cpplint.py --extensions=c,h my-program.c

Valgrind

- Valgrind is a programming tool for memory debugging, memory leak detection, and profiling.
- It's a simple tool which monitors every call to `malloc` (and other memory allocation functions) and `free` to make sure that all memory allocated to our program is subsequently released when we are done with it.

POSSIBLE MEM LEAK/MEMORY CORRUPTION SCENARIOS

- Use of uninitialised memory
- Reading/writing memory after it has been free'd
- Reading/writing off the end of malloc'd blocks
- Reading/writing inappropriate areas on the stack
- Memory leaks -- where pointers to malloc'd blocks are lost forever

```
valgrind --show-reachable=yes ./my-program  
"args"
```

```
valgrind --tool=memcheck --leak-check=yes --  
show-reachable=yes ./my-program
```